

ANALYTICAL PROBLEM-SOLVING ACTIVITY

TOPIC 1: METRICS FOR PERFORMANCE EVALUATION

1. ACCURACY

Given:

- Total samples = 100
- Correctly classified = 85

2. How many samples were classified incorrectly?

Incorrect = Total – Correct

$$\begin{aligned} &= 100 - 85 \\ &= \mathbf{15 \text{ samples}} \end{aligned}$$

3. Calculate the accuracy of the model.

Accuracy = (Correctly classified / Total) $\times$ 100

$$\begin{aligned} &= (85 / 100) \times 100 \\ &= \mathbf{85\%} \end{aligned}$$

4. Calculate the error rate.

Error Rate = (Incorrectly classified / Total) $\times$ 100

$$\begin{aligned} &= (15 / 100) \times 100 \\ &= \mathbf{15\%} \end{aligned}$$

5. Is the model performing well based on accuracy?

Yes. The model has **85% accuracy**, so it is performing reasonably well.

6. What happens to accuracy if 5 more samples are classified correctly?

New correct samples:

$$85 + 5 = \mathbf{90}$$

New accuracy:

$$(90 / 100) \times 100 = \mathbf{90\%}$$

Answer: Accuracy increases from 85% to 90%.

2. CONFUSION MATRIX

Given:

- $TP = 40$
- $TN = 35$
- $FP = 10$
- $FN = 15$

1. What is the total number of predictions?

$$\begin{aligned}\text{Total} &= TP + TN + FP + FN \\ &= 40 + 35 + 10 + 15 \\ &= \mathbf{100 \text{ predictions}}\end{aligned}$$

2. Calculate the accuracy.

$$\begin{aligned}\text{Accuracy} &= (TP + TN) / \text{Total} \times 100 \\ &= (40 + 35) / 100 \times 100 \\ &= 75 / 100 \times 100 \\ &= \mathbf{75\%}\end{aligned}$$

3. Calculate the error rate.

$$\begin{aligned}\text{Error Rate} &= (FP + FN) / \text{Total} \times 100 \\ &= (10 + 15) / 100 \times 100 \\ &= \mathbf{25\%}\end{aligned}$$

4. Which value represents incorrectly predicted positive cases?

$$\mathbf{FP \text{ (False Positive)} = 10}$$

5. Which value represents incorrectly predicted negative cases?

$$\mathbf{FN \text{ (False Negative)} = 15}$$

3. PRECISION AND RECALL

Given:

- $TP = 80$
- $FP = 20$
- $FN = 10$
- $TN = 90$

1. Calculate Precision

Formula:

$$\begin{aligned}\text{Precision} &= TP / (TP + FP) \\ &= 80 / (80 + 20) \\ &= 80 / 100 \\ &= \mathbf{80\%}\end{aligned}$$

2. Calculate Recall

Formula:

$$\begin{aligned}\text{Recall} &= TP / (TP + FN) \\ &= 80 / (80 + 10) \\ &= 80 / 90 \\ &= \mathbf{88.89\%}\end{aligned}$$

3. Which metric tells us how many predicted positive cases were actually positive?

Answer: Precision

4. Which metric tells us how many actual positive cases were correctly identified?

Answer: Recall

5. Which metric is more important when missing a positive case is dangerous?

Answer: Recall, because it measures how many actual positive cases are correctly identified.

4. COMPARE TWO CLASSIFIERS

Metric	Classifier A	Classifier B
Accuracy	90%	85%
Precision	80%	95%
Recall	70%	90%

Answers:

1. Which classifier has higher accuracy?

Classifier A (90%)

2. Which classifier has higher precision?

Classifier B (95%)

3. Which classifier has higher recall?

Classifier B (90%)

4. Which classifier would you choose if detecting as many positive cases as possible is important?

Classifier B, because it has a higher **recall (90%)**.

5. Give one reason why accuracy alone may not be enough.

Accuracy alone may be misleading when the dataset is **imbalanced**. Precision and recall give more information about the model's performance.

5. ANALYZE MODEL PERFORMANCE

Given:

- **TP = 70**
- **TN = 80**
- **FP = 20**
- **FN = 30**

1. **Accuracy** = $(TP + TN) / \text{Total}$
 $= (70 + 80) / 200$
 $= 150 / 200 = 0.75$ or 75%

2. **Precision** = $TP / (TP + FP)$
 $= 70 / (70 + 20)$
 $= 70 / 90 \approx 0.778$ or 77.8%

3. **Recall** = $TP / (TP + FN)$
 $= 70 / (70 + 30)$
 $= 70 / 100$
 $= 0.70$ or 70%

4. **Error Rate** = $(FP + FN) / \text{Total}$
 $= (20 + 30) / 200$
 $= 50 / 200$
 $= 0.25$ or 25%

5.Strength and Weakness:

strength: Precision is **77.78%**, meaning the model is fairly good at correctly identifying positive predictions.

Weakness : Recall is only **70%**, meaning the model misses about **30% of actual positive cases**.

TOPIC 2: PREDICTION ALGORITHMS

1. SIMPLE LINEAR PREDICTION

Given Data:

Advertising Cost (₹)	Sales
10	20
20	40
30	60
40	80

The relationship is: **Sales=2×Advertising Cost**

Answers

1. Identify the relationship between advertising cost and sales.

Answer: There is a **positive linear relationship**. Sales increase by ₹2 for every ₹1 increase in advertising cost.

2. Predict sales when advertising cost is ₹50.

Answer: Sales=2(50)=100

100 sales

3. What happens to sales when advertising cost increases by ₹10?

Answer: Sales increase by **20 units**.

5. Is this an example of classification or prediction?

Answer: It is a **prediction (regression) problem**.

5. Why is a linear prediction algorithm suitable here?

Answer: Because advertising cost and sales have a **straight-line relationship**.

2. PREDICT TEMPERATURE

Day	Temperature
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1	25°C
2	26°C
3	27°C
4	28°C
5	29°C
6	30°C
7	31°C
8	32°C
9	33°C
10	34°C
11	35°C
12	36°C
13	37°C
14	38°C
15	39°C

The temperature increases by 1°C every day.

1. Predict the temperature on Day 5

Day 4 = 28°C

$$28+1=29\text{ °C}$$

Answer: 29°C

2. Predict the temperature on Day 10

Starting from Day 4: $28+(10-4)\times 1$

$$=28+6=34\text{ °C}$$

Answer: 34°C

3. Predicted temperature on Day 15

$$28+(15-4)\times 1$$

$$=28+11=39\text{ °C}$$

Answer: 39°C

4. How much does the temperature increase for every additional day?

- Day 1 → 25°C
- Day 2 → 26°C
- Day 3 → 27°C
- Day 4 → 28°C

Therefore, temperature increases by: 1°C per day

5. Is this a classification or prediction problem?

Answer: Prediction problem.

Because we are predicting a **numerical value (temperature)**.

3. HOUSE PRICE PREDICTION

Given Data:

Area (sq.ft)	Price (₹ lakhs)
1000	40
1500	60
2000	80

The relationship is: **Price=0.04×Area**

Answers

1. Identify the relationship between area and price.

Answer: There is a **positive linear relationship**. As area increases, house price increases.

2. Predict the price of a 2500 sq.ft house.

Answer: Price=0.04×2500

₹100 lakhs (₹1 crore)

3. What happens to the price when area increases by 500 sq.ft?

Answer: Price increases by **₹20 lakhs**.

4. Which type of algorithm can be used for this problem?

Answer: Linear Regression can be used.

5. Is this classification or numerical prediction?

Answer: It is **numerical prediction (regression)**.

4. COMPARE ACTUAL AND PREDICTED VALUES

Actual	Predicted	Error
50	48	2
60	65	5
70	68	2
80	85	5
90	88	2

Formula: **Error=|Actual–Predicted|**

Answers

1. Which prediction has the smallest error?

Answer: The smallest error is **2**. It occurs for **50 → 48, 70 → 68, and 90 → 88**.

2. Which prediction has the largest error?

Answer: The largest error is **5**. It occurs for **60 → 65 and 80 → 85**.

3. Calculate the error for each prediction.

- $50 - 48 = 2$
- $60 - 65 = 5$
- $70 - 68 = 2$
- $80 - 85 = 5$
- $90 - 88 = 2$

4. Is the model generally predicting close to the actual values?

Answer: Yes. The errors are small, ranging from **2 to 5**, so the predictions are generally close.

5. Why is the difference between actual and predicted values important?

Answer: It shows the **accuracy of the prediction model**. A smaller difference means better prediction.

5. CHOOSING A PREDICTION ALGORITHM

Given Applications

- A. Predicting tomorrow's temperature.
- B. Predicting the price of a house.
- C. Predicting the number of products sold next month.
- D. Predicting whether an email is spam or not.

Answers

1. Which applications are prediction problems?

Answer: A, B, and C are prediction/regression problems.

2. Which application is a classification problem?

Answer: D — Predicting whether an email is spam or not.

3. Which algorithm could be used for predicting house prices?

Answer: Linear Regression can be used.

4. Why is regression suitable for numerical prediction?

Answer: Regression predicts **continuous numerical values**, such as temperature, sales, and house prices.

5.. Give one difference between classification and prediction.

Classification	Prediction (Regression)
Predicts a category/class	Predicts a numerical value
Example: Spam or Not Spam	Example: House price

